

A review of uncertainty quantification and its applications in numerical simulation of scramjet combustor

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ABSTRACT

Engine design heavily relies on numerical simulation of engine environments, which greatly accelerates the design iteration and increases efficiency. Nonetheless, for a long time, the uncertainty associated with numerical modeling has been overlooked due to resource constraints. The advent of increased computing power in recent years has enabled designers to consider the critical importance of uncertainty quantification(UQ) for developing more robust and reliable designs. UQ methods can be broadly classified into uncertainty propagation and inverse problem. Given the central role of uncertainty propagation in the inverse problem, this review focuses on the development of propagation methods and categorizes them into three groups based on the representation of uncertainty. Bayesian methods are also mentioned to provide a complete picture of UQ. The review highlights the successful applications of UQ in simulating chemical kinetics, turbulence, and scramjet, and outlines potential UQ techniques for practically simulating high-dimensional, strong-nonlinear engine combustion processes in the future.

1. Introduction

Uncertainty quantification(UQ) is the rational process by which proximity between predictions and observations is characterized [1]. In traditional numerical simulations, deterministic conditions are assumed to accurately represent the actual environment, such as a combustor. However, due to the complex nature of these environments, a finite number of physical parameters are insufficient to recover all relevant information about the physical process. As a result, deterministic simulations may only provide preliminary results to support engine design. To overcome this, UQ provides a statistical approach to compensate for omissions by modeling uncertainties in physical parameters such as inflow, combustion model, and geometric description. This review will focus on UQ methods for simulating combustor environments and their potential applications in improving engine design and safety.

In the context of combustor simulations, uncertainties commonly arise from physical parameters such as inflow, combustion model, and geometric description. As a result, deterministic simulations may provide incomplete predictions, under the condition of many uncertainty sources, and overlook critical phenomena, such as the ‘unstart’ [2,3] and blow-off [4,5], which can significantly shorten the engine’s life and reduce the reliability and safety of aircraft.

In recent years, people realized the high-fidelity model is impossible used in engineering for now, such as DNS, detailed chemical mechanism and Eulerian–Lagrangian spray atomization model. So some researchers tried to use the combination of low-fidelity model and UQ method [6,7] to cover the engineering problem, and assume this is a more economical way. Many reviews [8–16] have introduced the development and application of UQ in their respective fields and attracted much attention. The NASA Langley Multidisciplinary UQ Challenge [17], proposed by the NASA Langley Research Center, try to make UQ more practical and receives enthusiastic response [18–20].

Uncertainty can be generally classified into two groups [17]. One is aleatory uncertainty due to the inherent variation of physical systems, and the other is epistemic uncertainty caused by a lack of knowledge. The aleatory uncertainty mainly exists in the parameters from measurement, like inflow or physical fuel properties and geometry description. The relationship between measurement error and uncertainty is introduced in Section 3. The epistemic uncertainty is again classified into two groups: One is the essential epistemic uncertainty, existed in the parameters of high-fidelity model, which we are really lack of knowledge. And the other one is the limited epistemic uncertainty, which exists in the parameters of low-fidelity models, and can be calibrated by a high-fidelity model. The limited epistemic uncertainty can

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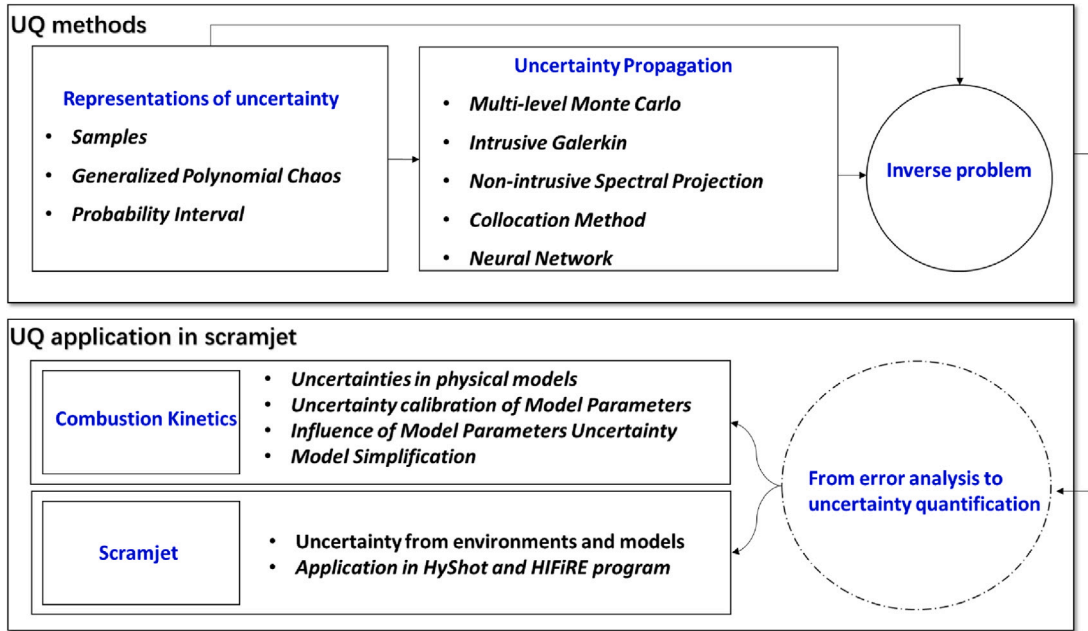


Fig. 1. The organization logic for this review.

even be removed by direct using the high-fidelity model. However, the high-fidelity model often requires unaffordable computing resources. As a result, the low-fidelity model is widely used, and the epistemic is unavoidable.

This article aims to introduce to engine designers the development and applications of UQ in scramjet design, especially combustion related topics. Similar work for other parts is ignored, such as the compressor, which can be referred to the work of Ghisu and Shahpar [21]. As Fig. 1 present for the organization logic, Section 2 mainly introduces uncertainty propagation methods based on the representation of uncertainty. In addition, the inverse problems are included. These methods aim at solving the high-dimensional, nonlinear problem in engine combustion robustly and efficiently. Section 3 try to bridging the traditional error analysis and the UQ approach. Section 4 focus on the source of uncertainties and applying UQ methods in combustion flow. Section 5 concludes this review, and give some suggestions for future work based on the review and some our experience.

2. UQ methods

This chapter will introduce UQ methods, focusing on the probabilistic class method, and at the same time, briefly introduce the non-probabilistic class method. Uncertainty is usually represented by probability, and the probabilistic method is based on the probability space $(\Omega, \mathcal{F}, P)$, where Ω is the sample space, representing the set of all possible events. From a perspective of mathematics, it is not easy to directly define probability in Ω , so the introduction of an algebraic structure $\mathcal{F}(\delta - \text{algebra})$ expresses a set of the subsets of Ω meeting certain conditions. P is the probability measure based on the algebraic structure $\mathcal{F}$. In essence, the probabilistic UQ method is how to effectively represent or propagate the probabilistic structure in space Ω and calculate its transformation through various mappings. However, the non-probabilistic methods consider that the limited data cannot effectively approach P in the probability space [22]. Therefore, by introducing more mathematical structures, P is blurred with probability interval. The construction process of the probabilistic and non-probabilistic approach illustrates that the connection between them is closely, and most UQ methods can be generalized for the two approaches.

Before introducing the specific UQ methods, the following mapping is first considered:

$$\mathbf{y} = M(\mathbf{x}, \boldsymbol{\theta}) \quad (1)$$

where $\mathbf{x}$ and $\boldsymbol{\theta}$ are the input vector and parameter vector, respectively. $\mathbf{y}$ is the output vector and M can represent any complex mapping in general. Specifically in this paper, the integration of various differential equations and algebraic equations in scramjet simulation is the total mapping. Ignoring some details of the entire system, it can be seen that it consists of input $\mathbf{x}$, output $\mathbf{y}$, and system parameters $\boldsymbol{\theta}$. The output vector $\mathbf{y} = \{y_1, y_2, \dots, y_n\}$ represents a series of the quantity of interest (QoI), and we can measure them in the experiment, such as pressure, temperature, composition, and other physical quantities. Input vector $\mathbf{x} = \{x_1, x_2, \dots, x_m\}$ includes a series of boundary conditions and initial conditions. System parameter $\boldsymbol{\theta} = \{\theta_1, \theta_2, \dots, \theta_p\}$ mainly comes from the parameters of various mathematical models in the system. Since the latter two have random and unavoidable uncertainties, the output will inherit the uncertainty from the mapping system. The task of UQ is to determine the probabilistic structure of input and parameter and calculate the probabilistic structure of output results after mapping, or vice versa.

2.1. Representations of uncertainty

In order to facilitate calculation, an uncertainty representation method should be considered first. The probability density function (pdf) $f(\xi)$ is the most natural choice, which can intuitively reflect the probability size in different regions in probability space. In engineering, statistic characteristics such as mean value and variance can be obtained by the moment integral formula of the pdf, where the mean value is the first central moment, and the variance is the second central moment. There exist two other representations of uncertainty: the cdf(cdf) and the characteristic function(cf), which are the integrated form of pdf and the Fourier transform of pdf, respectively. It should be noted that there are slight differences in the actual application of different uncertainty representation methods. Seshadri et al. [23] give the measured distance of two pdfs, which can be used for objective optimization. Cook&Jarrett [24] extended a similar idea to the cdf and proved that the use of the cdf could have a lower computational cost than the use of the pdf in the optimization problem. In addition, it

should be noted that the function of cdf and cf always exist in the probability space, while the pdf does not, such as the δ -pdf function.

It can be seen from the above analysis that the pdf is at the center of the uncertainty representation method. However, the pdf is generally not easy to obtain, or even cannot be expressed in algebraic form. Therefore, different approximation methods are the core problem of representing uncertainty in the actual calculation process.

2.1.1. Samples

The samples obtained by repeated sampling from the pdf is the most straightforward representation of uncertainty. As long as the sampling method is reasonable and correct, the more samples, the more details of the pdf can be obtained. The performance of various UQ methods based on the Monte Carlo(MC) method is directly and closely related to the quality and quantity of samples. At the same time, the least square method can be used to fit the pdf.

Some other sampling methods are available when direct sampling is difficult, including accept–reject sampling, importance sampling, and Metropolis–Hastings(MH) sampling. Liu [25] compared the three sampling methods and found that the sample quality of accept–reject sampling was the best, the MH algorithm was second, and importance sampling was last. However, accept–reject sampling requires knowing the envelope constant of the pdf in advance, while the MH sampling does not, so the MH algorithm is optimal by comprehensive comparison.

MH algorithm is based on the construction of the Markov Chain. For the discrete random sequence $x^{(i)} \in \mathcal{X} = \{x_1, x_2, \dots, x_s\}$, if the condition $p(x^{(i)}|x^{(i-1)}, \dots, x^{(1)}) = T(x^{(i)}|x^{(i-1)})$ is satisfied, that is, the development of the random sequence in space $\mathcal{X}$ is only determined by the current state and a fixed transition matrix, then the whole process is called the Markov chain. When the transformation matrix T meets the following two conditions, the entire Markov chain has a stable state:

- *Irreducible*, that is, all states have a positive probability, that is, the transition matrix cannot be decomposed into the smaller independent matrix
- *Aperiodicity*, the whole conversion chain will not fall into an endless cycle.

For example, if the discrete sequence is $\mathcal{X} = \{x_1, x_2, x_3\}$, its probability distribution is $p(\mathcal{X}) = \{p(x_1), p(x_2), p(x_3)\}$, and the transition probability matrix is $T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0.1 & 0.9 \\ 0.6 & 0.4 & 0 \end{bmatrix}$ [26]. So no matter what the initial probability $p_0(\mathcal{X})$ is, it converges to $p_\infty(\mathcal{X}) = p_0(\mathcal{X})T^\infty = (0.2213, 0.4098, 0.3689)$ after an infinite number of transitions. This convergence property can be applied to distributions where sampling is difficult. A sufficient and non-necessary condition for existence of $p_\infty(\mathcal{X})$ is:

$$p_\infty(x^{(i)})T(x^{(i-1)}|x^{(i)}) = p_\infty(x^{(i-1)})T(x^{(i)}|x^{(i-1)}). \quad (2)$$

This condition is also known as the detailed balance condition.

The MH algorithm aims to design such a transfer function so that the probability distribution function can converge. First, let the target distribution be recorded as $p(x)$. The wandering distribution function $q(x^*|x)$ is used to calculate the current state x the probability of the next state x' conversion. The transfer chain composed of $p(x)$ and $q(x^*|x)$ generally does not satisfy the fine equilibrium condition $p(x)q(x^*|x) \neq p(x^*)q(x|x^*)$. The MH algorithm introduces the acceptance and rejection function $A(x, x^*) = \min\{1, \frac{p(x^*)q(x|x^*)}{p(x)q(x^*|x)}\}$. And then, a continuous form of the transformation matrix T can be written:

$$K_{MH}(x^{(i+1)}|x^{(i)}) = q(x^{(i+1)}|x^{(i)})A(x^{(i)}, x^{(i+1)}) + \delta_{x^{(i)}}(x^{(i+1)})r(x^{(i)}) \quad (3)$$

In that rejection function

$$r(x^{(i)}) = \int_{\mathcal{X}} q(x^*|x^{(i)})(1 - A(x^{(i)}, x^*))dx^*.$$

It is not difficult to prove that $p(x)$ and transfer function $K_{MH}(x^{(i+1)}|x^{(i)})$ constitute a Markov chain that satisfies the detailed balance condition. Based on the above construction, complicated distribution $p(x)$ can be sampled at the stable state of the Markov chain.

Whether the whole MH algorithm sample is good, bad, or failing is closely related to the choice of $q(x^*|x)$. $q(x^*|x)$ controls the conversion probability of different samples x points to the next sampling point. Common choices include normal form $q(x^*|x) = \mathcal{N}(x, \sigma^2)$ and independent sampling process $q(x^*|x) = q(x^*)$.

The MH algorithm and its various variant forms are the cornerstones of the Markov chain Monte Carlo method (MCMC), while the MCMC is a standard method for parameter uncertainty calibration. This is because the pdf of the model parameters is usually not a simple distribution function, and the regularity may also be difficult to guarantee, so using the sample to represent its uncertainty is a more robust method. The MCMC method can provide samples of these complex distributions. The sample has been widely used in various model parameter calibration methods [27,28].

2.1.2. General polynomial chaos

From the perspective of the function space, the pdf can be regarded as a vector in the second-order probability space $(\Omega, \mathcal{F}, P)$, so the form of coordinates and basis functions can yield:

$$f(\xi) \approx \sum_{\alpha \in I} \hat{f}_\alpha \phi_\alpha(\xi) \quad (4)$$

If this expansion can achieve spectral accuracy, it can also be called spectral expansion [29]. So the construction of the basis function is the key to the problem. Karhunen–Loève is one of the commonly used methods for constructing basis functions. It is the optimal construction method of basis functions in the sense of mean square norm [11]. However, its construction process is complicated and computationally intensive, so it is difficult to be promoted. Wiener [30] notices that the Hermite polynomial is the best basis function for Gaussian distribution and the polynomial chaos expansion(PCE) is built. Cameron&Martin [31] proves that the Hermite-polynomial has an approximation ability for arbitrary distribution, but the convergence speed for different distributions are different. Xiu&Karniadakis [32] uses the Askey algorithm to generalize Wiener's choice of polynomial, proposes the generalized PCE(gPCE). They choose different polynomials for different distributions to obtain the exponential convergence of approximation ability. Because the various operations of polynomial are quite cheap, compared to the MC method. So it has been rapidly popularized. Moreover, as a basis function in the second-order probability space, it satisfies the unit orthogonality:

$$\langle \phi_\alpha, \phi_\beta \rangle_{(\Omega, \mathcal{F}, P)} = \int_{\Omega} \phi_\alpha(\xi) \phi_\beta(\xi) dP(\xi) = \delta_{\alpha, \beta} \quad (5)$$

The multiple indicator $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_d\}$ indicates that the multi-dimensional polynomial basis function ϕ_α is a d -dimensional function composed of α_i -order one-dimensional polynomial basis functions in each dimension $\phi_\alpha = \phi_\alpha(\phi_{\alpha_1}^1, \phi_{\alpha_2}^2, \dots, \phi_{\alpha_d}^d)$. The one-dimensional polynomial basis function has its optimal choice for different probability spaces. The uniform distribution corresponds to the Legendre polynomial, the Gaussian distribution corresponds to the Hermite polynomial, and the beta distribution corresponds to the Jacobi polynomial.

The indice of α will traverse the indice set I , so the choice of I will affect the approximation of the left and right sides of Eq. (4). Tang&Zhou [11] gives three alternative indicators set construction methods. The complete polynomial spatial index set $I_{d,k}^T$, tensor polynomial spatial index set $I_{d,k}^P$ and hyperbolic polynomial spatial indicator set $I_{d,k}^H$

$$I_{d,k}^T = \{\alpha \in \mathbb{N}^d \mid |\alpha|_\infty \leq k\}$$

$$I_{d,k}^P = \{\alpha \in \mathbb{N}^d \mid |\alpha|_1 \leq k\}$$

$$I_{d,k}^H = \{\alpha \in \mathbb{N}^d \mid \prod_j \alpha_j + 1 \leq k + 1\}$$

The degrees of freedom(DOF) of each indice set corresponding to spanning space are $\dim^T = (k+1)^d$, $\dim^P = \binom{k+d}{k}$, $\dim^H \leq \lfloor (k+1)^d (1 + \log(k+1))^{d-1} \rfloor$. Considering the balance between spatial freedom and precision of spanning space, the $I_{d,k}^P$ indice set is generally chosen as the form of high-dimensional space. However, in addition to DOF and precision, the choice of indice set can have additional effects on calculating equations. Pulch&Xiu [33] constructed an example to prove that, for the Galerkin discrete equations of the linear conservation equation, the indice set $I_{d,k}^P$ makes the reconstruction equations of the example no longer maintain hyperbolic properties, while $I_{d,k}^T$ does not have this disadvantages.

2.1.3. Probability interval

In the actual application process of the project, since the available data is usually not enough to obtain an accurate pdf [34], a natural idea is to find a probability interval composed of the upper and lower bound, and to guarantee the accurate pdf falls between the two bounds. It is important to note that there is a difference between probability intervals and the uncertainty intervals. The latter essentially refers to the upper and lower bounds of a closed interval in the probability space. According to the generalized PCE, there is always an optimal polynomial approximation to represent the probability interval. The definition of the probability interval comes from several standard non-probability UQ methods, including evidence theory [22], fuzzy set [35], and probability box [36].

The evidence theory is one of this methods. It introduces the evidence space $(\Omega, \mathcal{F}, m)$ as its theoretical base. The evidence space is the result of replacing the probability measure in the probability space $(\Omega, \mathcal{F}, P)$ with the belief mass m . The primary belief distribution function is defined $\sum_{A \in \mathcal{F}} m(A) = 1$. Based on the definition, the belief quality of event A is not the same as the probability measure, where the latter one has the property $P(A = \Omega) = 1$. The upper bound of pdf is defined as plausibility, and the lower bound is defined as belief. Belief represents the sum of belief mass of all subsets of A . Plausibility represents the sum of belief mass of all sets containing A .

$$Bel(A) = \sum_{B|B \subseteq A} m(B), \quad Pl(A) = \sum_{B|B \cap A \neq \emptyset} m(B) \quad (7)$$

For the above two measures, the following relationship hold:

$$\begin{aligned} Bel(A) + Pl(A^c) &= 1 \\ Bel(A) + Bel(A^c) &\leq 1 \\ Pl(A) + Pl(A^c) &\geq 1 \end{aligned} \quad (8)$$

Then we can further define the cumulative belief function(CBF) and the cumulative plausibility function(CPF).

$$\begin{aligned} CBF &= \{[x, Bel(\mathcal{A}_x)] : x \in \Omega\} \\ CPF &= \{[x, Pl(\mathcal{A}_x)] : x \in \Omega\} \\ \mathcal{A}_x &= \{\tilde{x} : \tilde{x} \in \Omega \text{ and } \tilde{x} \leq x\} \end{aligned} \quad (9)$$

Probability, belief, and plausibility have the following relationship for the consistent probability space and evidence space.

$$Bel(A) \leq P(A) \leq Pl(A) \quad (10)$$

Therefore, the relationship between the CDF, CBF, CPF is shown in Fig. 2. This converts the representation of pdf to the belief and plausibility functions. Actually, the representations of the function of CBF and CPF can still be used with samples and spectral expansion, as CPF.

2.2. Uncertainty propagation

Uncertainty propagation is the core of UQ methods. Besides of its own function to compute the output uncertainty, it is also the kernel of inverse problem. This is because, generally speaking, the properties

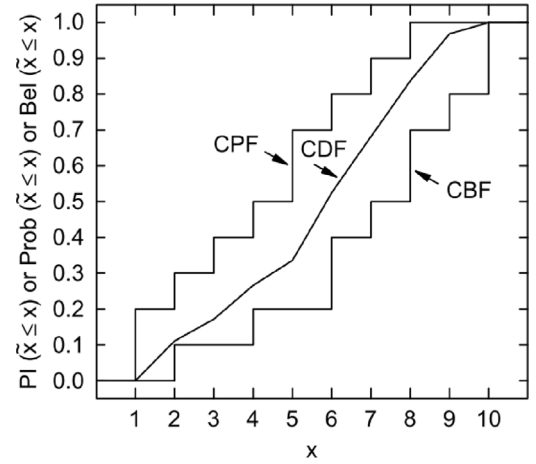


Fig. 2. The relationship between CPF, CBF and CDF [37].

of inverse mapping in Eq. (1) may be very poor, and there is no simple representation based on the existing mathematical tools. The direct calibration of inputs and parameters is not possible, and the inverse problem always fall into an optimization problem or Bayesian problem, where the uncertainty propagation would be computed repeatedly. Therefore, we mainly reviewed the methods of uncertainty propagation.

2.2.1. Multi-level Monte Carlo

The multi-level monte carlo(MLMC) is one of the acceleration methods of the MC method, which needs a huge amount of samples [38]. It was first used to solve the spatial discrete function integral problem [39], and was later introduced into the stochastic differential equation [40], elliptic partial differential equation(PDE) [41] and The hyperbolic partial differential equation [42]. It is one of the most attracting UQ methods. By constructing different levels of meshes, that is, sampling on coarsening meshes, one can quickly obtain the statistic information under the finest mesh with a small number of samples.

Referring to [43], a brief introduction to the construction process of the MLMC method is given. Let the mean and variance of the probability space $(\Omega, \mathcal{F}, P)$ with respect to the function f be $\mathbb{E}(f)$ and $\mathbb{V}(f)$, respectively, then the traditional MC method defines an unbiased sample estimate for $\mathbb{E}(f)$ as

$$E_N(f) = N^{-1} \sum_{i=1}^N f(\xi^i) \quad (11)$$

Where ξ^i samples from the same probability space. According to the central limit theorem, the variance of the estimate is easily obtained as $V(E_N) = N^{-1} \mathbb{V}(f)$, so the convergence speed of the MC method error is $\epsilon \sim O(N^{-\frac{1}{2}})$. Therefore, in order to obtain an accurate estimate, a large number of samples are required, which is unfriendly for some single-sample expensive calculations.

A method of acceleration is the MLMC method, for which a multi-level estimate is first designed.

$$G_{ml} = \sum_{\ell=0}^L G_{\ell}, \quad G_{\ell} = N_{\ell}^{-1} \sum_{i=1}^{N_{\ell}} (f_{\ell}^{(\ell,i)} - f_{\ell-1}^{(\ell,i)}) \quad (12)$$

If $f_{-1} = 0$, it is not difficult to prove the following properties for multi-level estimate of G_{ml} .

$$\mathbb{E}(G_{ml}) = \mathbb{E}(f_L), \quad \mathbb{V}(G_{ml}) = \sum_{\ell=0}^L N_{\ell}^{-1} V_{\ell}, \quad V_{\ell} \equiv \mathbb{V}(f_{\ell} - f_{\ell-1}) \quad (13)$$

The error between the multi-level estimate and the true mean is defined by the mean square error:

$$\epsilon^2 = \mathbb{E}((G_{ml} - \mathbb{E}(f))^2) = \mathbb{V}(G_{ml}) + (\mathbb{E}(G_{ml}) - \mathbb{E}(f))^2 \quad (14)$$

Where f is the convergence state of the function sequence f_ℓ , that is $G_\infty - \mathbb{E}(f) = 0$. In order to enable the error to be less than a given threshold $\epsilon^2 < \epsilon_0^2$, it is generally assumed that the error is divided into two parts of equal parts, namely $\mathbb{V}(G_{ml}) < \epsilon_0^2/2$, $(\mathbb{E}(G_{ml}) - \mathbb{E}(f))^2 < \epsilon_0^2/2$. But it should be pointed out that this splitting method is not always optimal.

The principle of the MLMC method to obtain the acceleration effect is to estimate G_ℓ through a reasonable structural level, to ensure that the estimated variance decreases continuously as the level increases, and finally lead to using fewer samples on highest level. According to this process, the whole time cost is reduced. The usual hierarchical construction is geometric, that is, the different mesh densities. The MLMC method means that a small number of samples can be used on a denser mesh, while more samples are placed on a coarse mesh to achieve the desired error and cost balance. For example, let C_ℓ and V_ℓ be the computational cost of one sample in level ℓ and the variance of G_ℓ , respectively. Then the cost and variance of G_{ml} for multi-level estimation can be written as:

$$C_{ml} = \sum_{\ell=0}^L N_\ell C_\ell, \quad V_{ml} = \sum_{\ell=0}^L N_\ell^{-1} V_\ell \quad (15)$$

In order to reduce the variance of the global calculation, consider the problem of linear combination optimization with cost:

$$\min \sum_{\ell=0}^L (N_\ell^{-1} V_\ell + \mu^2 N_\ell C_\ell) \quad (16)$$

μ is Lagrange multiplier, Then the optimal sampling Numbers at different levels can be obtained by using the inequality relation $N_\ell = \mu \sqrt{V_\ell / C_\ell}$. To determine the error condition $V_{ml} = \epsilon_0^2$, we can get Lagrange multiplier $\mu = \epsilon_0^{-2} \sum_{\ell=0}^L \sqrt{V_\ell C_\ell}$. So the total cost can be written as

$$C_{ml} = \epsilon_0^{-2} \left(\sum_{\ell=0}^L \sqrt{V_\ell C_\ell} \right)^2 \quad (17)$$

And the cost of the traditional MC method can be written as $C = \epsilon_0^{-2} V_0 C_L$. It means Through the reasonable arrangement of C_ℓ and V_ℓ hierarchy, enables $C_{ml} < C$.

It should be pointed out that the hierarchical structure of MLMC method does not only depend on the mesh density. Its core idea fall on the cost and error variation of different levels. Geraci et al. [44] proposed multi-fidelity monte carlo(MFMC) method using different fidelity models to construct the hierarchy. This concept of multi-fidelity is the generalized version of multi-level, and the geometric MFMC method can further improve the computational efficiency.

2.2.2. Intrusive Galerkin

By using the PCE, the idea of intrusive Galerkin method is to use Galerkin method to project the governing equations into the polynomial basis space with a spectral expansion. So one can rewrite the governing equation into a set of spectral coefficient equations. Then the uncertainty propagation of probability space can be described by the evolution process of spectral coefficient. This method has been successfully applied to elliptic PDEs [32] and Boltzmann equations [45]. Considering the flow in the combustion chamber, the governing equation is a nonlinear hyperbolic Navier–Stokes equation with source term

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^{\text{inv}}(\mathbf{U})}{\partial \mathbf{x}} = \frac{\partial \mathbf{F}^{\text{vis}}(\mathbf{U})}{\partial \mathbf{x}} + \mathbf{S}^{\text{chem}} \quad (18)$$

The turbulence and combustion turbulence caused by the mesh scale are ignored here. Considering the finite projection of the conservation Eq. (18) in the probability space:

$$\left\langle \frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^{\text{inv}}(\mathbf{U})}{\partial \mathbf{x}}, \phi_k \right\rangle = \left\langle \frac{\partial \mathbf{F}^{\text{vis}}(\mathbf{U})}{\partial \mathbf{x}} + \mathbf{S}^{\text{chem}}, \phi_k \right\rangle, \quad k = 0, \dots, P \quad (19)$$

In order to calculate, the finite spectrum of conserved quantities, fluxes and source terms is expanded (20) into the projection process (19).

$$\begin{aligned} \mathbf{U} &\approx \sum_{j=0}^P \hat{\mathbf{U}}_j \phi_j & \mathbf{F}^{\text{inv}} &\approx \sum_{j=0}^P \hat{\mathbf{F}}_j^{\text{inv}} \phi_j \\ \mathbf{F}^{\text{vis}} &\approx \sum_{j=0}^P \hat{\mathbf{F}}_j^{\text{vis}} \phi_j & \mathbf{S}^{\text{chem}} &\approx \sum_{j=0}^P \hat{\mathbf{S}}_j^{\text{chem}} \phi_j \end{aligned} \quad (20)$$

Using the unit orthogonality of the basis function (5), the following spectral coefficient equations can be obtained:

$$\frac{\partial \hat{\mathbf{U}}_j}{\partial t} + \frac{\partial \hat{\mathbf{F}}_j^{\text{inv}}(\mathbf{U})}{\partial \mathbf{x}} = \frac{\partial \hat{\mathbf{F}}_j^{\text{vis}}(\mathbf{U})}{\partial \mathbf{x}} + \hat{\mathbf{S}}_j^{\text{chem}}, \quad j = 0, \dots, P \quad (21)$$

If the DOF in $\mathbf{U}$ itself is d , then the system (21) contains a total of $d \times (P+1)$ PDEs. Although the reformulated system of equations is huge, Elma et al. [46] reported a preconditioner technique to accelerate the linear system solving, and achieve a faster UQ approach compared with collocation method.

For the nonlinear hyperbolic equations, Després et al. [47] notes that its truncated Eqs. (21) do not guarantee hyperbolic properties, which means that it is difficult to guarantee the stability of a series of numerical solution tools developed based on hyperbolic conservation equations. Therefore, he proposed that the entropy variable [48] could be used to replace the conserved variable as the expansion variable, so as to avoid the problem of hyperbolic loss. Pettersson et al. [49] uses Roe variable to solve hyperbolic problem and obtain faster speed. Wu et al. [50] solved the stability problem by using the quasi-linear equation and non-conserved flux, in order to avoid the resource consumption in the process of variable substitution in the above two schemes. However, the solution of non-conservation form equation is not stable [51], so the practical application of this method needs to be discussed. It should also be noted that the discontinuity in the hyperbolic equation will affect the convergence of Galerkin's method. Jin&Ma [52] uses the discrete form of hyperbolic equation and the regularized numerical flux to construct the solution scheme of linear hyperbolic equation with discontinuous coefficient, and obtains the convergence property of spectral accuracy. However, it is still difficult to extend this idea to more general nonlinear hyperbolic equations with discontinuities.

Debusschere et al. [53] introduced the intrusive Galerkin method to chemical reactions. Reagan et al. [54] designed a quick exponential logarithm algorithm based on Hermite polynomial in order to make the calculation of uncertainty propagation of kinetic model faster. They used the quasi-spectral process to calculate the spectral coefficients in chemical reactions. However, because of the strong nonlinear nature of chemical reaction, the uncertainty deviates from the gaussian distribution in the propagation process, so it is difficult to obtain a good accuracy by using Hermite polynomial approximation alone. While for non-gaussian distribution, the rapid calculation of exponential logarithm is difficult to be generalized. In addition, it should be noted that global polynomials are not friendly to strong non-regular in chemical reactions and are easy to cause divergence problems. Therefore, LeMaitre et al. [55,56] introduced the basis function of Haar wavelet, which can better represent the propagation process of uncertainty in nonlinear systems.

The intrusive Galerkin method requires rewriting the governing equations, which is a great obstacle to the application of the algorithm to engineering. Moreover, for the hyperbolic non-linear equations with chemical reaction source term, the existing mathematical formula cannot guarantee the stability of the numerical simulation. So it is difficult to extend the intrusive Galerkin to the actual calculation of scramjet.

2.2.3. Non-intrusive spectral projection

Although it is difficult to generalize intrusive Galerkin to real computation, we still hope to obtain the property of rapid convergence of spectral expansion. Combining the idea of response surface and PCE, a natural idea is to directly construct the polynomial surrogate of the quantity y in the probability space in the mapping system (1). The spectral coefficients of polynomial surrogate can be calculated with samples, and thus avoid the reformulating equations. This is the basic idea of Non-intrusive spectral projection (NISP) method [57–59]

$$y(\xi) = \sum_{i=0}^N \hat{y}_i \phi_i(\xi), \quad \xi \in (\Omega, \mathcal{F}, P) \quad (22)$$

Reagan et al. [59] assumed that y was consistent with or close to the gaussian distribution, and chose Hermite polynomial as the basis function. However, in real complex systems, the specific quantities concerned are varied in a wide range. The shape of each y is likely to deviate from the gaussian distribution. So the single Hermite basis may be difficult to obtain true spectral accuracy. Immediately the gPCE [32] can effectively remedy this shortcoming.

The spectral coefficient can be obtained by using the inner product projection in the probability space, which depends on the integral technique in numerical calculation:

$$\hat{y}_i = \int_{\Omega} y(\xi) \phi_i(\xi) \pi(\xi) d\xi = \int_{\Omega} f(x(\xi), \theta(\xi)) \phi_i(\xi) \pi(\xi) d\xi \quad (23)$$

Corresponding to the density function π of different probability Spaces, there are corresponding optimal gaussian quadrature rules, such as gauss-hermite quadrature and gauss-legendre quadrature corresponding to gaussian distribution and uniform distribution respectively. Therefore, we only need to calculate the “sample” at the gaussian quadrature points, and can quickly solve the coefficient (23), and then obtain the statistic information. However, it should be noted that the optimal gaussian quadrature rule is established on the basis that the mapping $f(x, \theta)$ is a regular mapping. However, the working environment in the scramjet is difficult to ensure such a nice mapping. So some non-optimal gaussian quadrature rules will achieve similar quadrature accuracy [60] with a small needed samples, such as the Clenshaw–Curtis quadrature.

2.2.4. Collocation method

NISP method is an efficient UQ method, but it needs a good estimation of the probability shape of the mapping result y to obtain a high precision. Another troublesome problem is that the choice of polynomial chaos basis functions is difficult to gain any advantage if y is far away from all the commonly used known probability shapes. Similar to Eq. (22), a more general surrogate construction method can choose Lagrangian interpolation [61], namely, to give up spectral accuracy and obtain the advantage of flexible collocation points.

$$f(\xi) \approx Q(f)(\xi) = \sum_{k=1}^N f(\xi_k) \mathcal{L}_k(\xi) \quad (24)$$

For N collocation points, Lagrangian interpolation polynomial needs to satisfy property $\mathcal{L}_k(\xi_j) = \delta_{kj}$. A general construction form is as follows:

$$\mathcal{L}_k(\xi) = \prod_{\substack{j=1 \\ j \neq k}}^N \frac{\xi - \xi_j}{\xi_k - \xi_j} \quad (25)$$

The choice of collocation point is the key of calculation. Although Lagrangian interpolation theoretically guarantees that any selected point can be calculated, some good interpolation points can have better accuracy and other advantages. For example, under the condition of $\xi \in [-1, 1]^d$, the selection of Clenshaw–Curtis points as interpolation points can guarantee the optimization of high-dimensional sparse grid [61]. As the dimension of the calculated probability space increases, the multi-dimensional interpolation points constructed in the

form of complete tensors is power exponential $O(N^d)$. This rate of growth, also known as the “curse of dimension”, is unbearable for any computing resources. Sparse mesh is an accelerated numerical technique for high-dimensional interpolation quadrature, which can reduce the growth rate of sample points of multi-dimensional interpolation to $O(N(\log N)^{d-1})$ [62], and is one of the methods to alleviate the “curse of dimensions”. The construction process of Smolyak method is introduced here.

Remember the one-dimensional interpolation form of the i th layer is as follows

$$Q^i(f) \equiv \sum_{k=1}^{N_i} f(\xi_k^i) \mathcal{L}_k^i(\xi^i) \quad (26)$$

Then the sparse grid first attempts to construct a multi-layer difference interpolation form

$$\begin{aligned} \Delta_l^i(f) &\equiv (Q_l^i - Q_{l-1}^i)(f), \quad l \geq 1 \\ Q_0^i(f) &\equiv 0 \end{aligned} \quad (27)$$

Two requirements of the collocation points should be considered for each layer: Firstly with increasing layer number, the points at $l-1$ layer should be included by l layer as much as possible, which can significantly decrease the final total number of points. Secondly, the selected points at each layer should make the accuracy of its interpolation meaningful. For example the Gaussian quadrature rule is the most accurate interpolation rule [63]. Clenshaw–Curtis rule is another one, but not accurate as Gaussian quadrature rule, where the former only reaches N -order accuracy with N points and the latter can reach $(2N-1)$ -order accuracy. But it is nested best, and can make the multi-level structure have the least number of points. So Clenshaw–Curtis is generally suggested in sparse grid and read

$$\xi_j = -\cos \frac{\pi(j-1)}{N^k-1}, \quad j = 1, \dots, N^k \quad (28)$$

where the index $k > 1$ represents the level. The multi-level selection points based on Clenshaw–Curtis rule can be represented as $N^k = 2^{k-1} + 1$. At the same time, it can be predicted that a reasonable one-dimensional interpolation method can ensure that the difference interpolation will increase with the increase of the level, and its limit is close to zero, $\Delta_{l \rightarrow \infty}^i \rightarrow 0$. The multi-dimensional interpolation writing of Smolyak sparse grid:

$$Q_{\mathcal{L}}^d(f) = \sum_{\|\mathbf{l}\|_1 \leq d+\mathcal{L}} (\Delta_{l_1}^1 \otimes \dots \otimes \Delta_{l_d}^d)(f) \quad (29)$$

where $\otimes$ represents the tensor product and $\mathbf{l} = (l_1, \dots, l_d) \in \mathbb{N}^d$, $\mathcal{L} \geq 0$ represents the approximation degree of the sparse grid. At the same time, difference interpolation can also be used to represent the multi-dimensional interpolation form of the complete tensor [29].

$$Q_{\mathcal{L}}^d(f) = \sum_{\|\mathbf{l}\|_{\infty} \leq d+\mathcal{L}} (\Delta_{l_1}^1 \otimes \dots \otimes \Delta_{l_d}^d)(f) \quad (30)$$

where $Q_{\mathcal{L}}^d$ represents the cubature operator, $\mathcal{L}$ represents the maximum level for each quadrature dimension, and d represents the total dimension. Upon examining Fig. 3, it is found that the Smolyak structure reduces the range of multidimensional index from infinite norm to 1 norm. Its essence is to utilize the attenuation characteristic of the difference interpolation (27) in order to get rid of the less important points. In this way, the sparse grid is built from the complete tensor construction. Also Eq. (29) can be written in the form of non-difference interpolation [64] to facilitate the construction of numerical method.

$$Q_{\mathcal{L}}^d(f) = \sum_{\substack{\max\{d, \mathcal{L}+1\} \leq \|\mathbf{l}\|_1 \leq d+\mathcal{L} \\ \mathbf{l} \in \mathbb{N}^d, l_i \geq 1}} (-1)^{\mathcal{L}+d-\|\mathbf{l}\|_1} \binom{d-1}{\mathcal{L}+d-\|\mathbf{l}\|_1} \bigotimes_{i=1}^d Q_{l_i}^i(f) \quad (31)$$

where $Q_{l_i}^i$ denote the quadrature in dimension i , and $\binom{n}{k}$ is the combination operator. The other symbols are the same as Eq. (30).

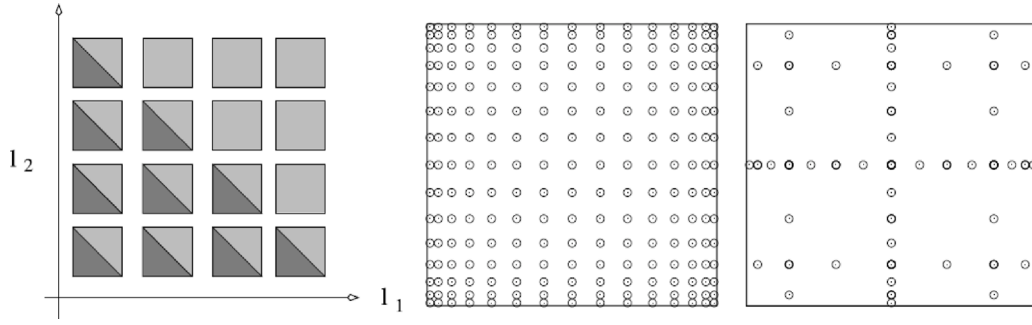


Fig. 3. Sparse Grid based on Fejér rule, $d = 2$, $L = 4$. Left: the diagram of indice in Eq. (30). The square is tensor grid and the triangle is Smolyak grid; Middle: points of tensor grid at domain $\{(-1, 1) \times (-1, 1)\}$; Right: points of Smolyak grid at domain $\{(-1, 1) \times (-1, 1)\}$ [29].

2.2.5. NN surrogate

More recently, Scientific machine learning (sciML) is emerging in the science community. It aims to use the machine learning (ML) tools and the scientific data sets to extract the physics of the complex systems. Among the various ML tools, the neural network (NN) of solving differential equations is the most attractive topic. DNN [65], PINN [66], TgNN [67] and DeepONets [68] are developed to be used as a surrogate of differential equations, which is the core mapping in CFD. The usage of various NNs in uncertainty is quite simple: once the networks are constructed successfully, it can be used as a surrogate, and produce any amount of samples as we want. So the native MC method can immediately be applied on this surrogate to computing the uncertainty propagation. So the main question is how to construct a successful network. In this chapter, we review the networks which have been applied in UQ region, and the other networks are ignored.

DNN is the most simple neural network, which can utilize the ideal of shortcut in ResNet [69] to construct a deep enough network by inputs and outputs. The network is composed by multi-layer structure:

$$\begin{aligned} \mathbf{X}^{l+1} &= \sigma(\mathbf{W}^l \mathbf{X}^l + \mathbf{B}^l) \\ \hat{\mathbf{Y}} &= \sigma^t(\mathbf{W}^{L-1} \mathbf{X}^{L-1} + \mathbf{B}^{L-1}) \end{aligned} \quad (32)$$

where the $\mathbf{X}^l$ is the input of layer l and $\mathbf{W}^l$ and $\mathbf{B}^l$ are the linear coefficient and bias of connection of layer l and layer $l + 1$. Then the output of layer l is obtained with a non-linear function σ and is used as the input of layer $l + 1$. The non-linear function is the activation function, and its popular choice includes the tangent function, rectified linear unit (ReLU) function and Swish function.

$$\begin{aligned} \sigma(x) &= \tanh(x) && \text{tanh function} \\ \sigma(x) &= \max(0, x) && \text{ReLU function} \\ \sigma(x) &= \frac{x}{1 + \exp(-\gamma x)} && \text{Swish function} \end{aligned}$$

Generally Swish function is preferred with $\gamma = 1$. At last, the final output $\hat{\mathbf{Y}}$ is obtained by the task operator σ^t

$$\sigma^t \begin{cases} \text{sigmoid}(x) & \text{classification task} \\ 1 & \text{regression task} \end{cases} \quad (33)$$

and the sigmoid function is read

$$\text{sigmoid}(x) = \frac{1}{1 + \exp(x)} \quad (34)$$

In most situations, the output of differential equations are continuous function, and the training is a regression task. However, we can set whether the flame is blow off as the output, and then the training of the NN is the classification task.

The shortcut is the additive connection of layer l and layer $l + k$, where $k > 1$. And it is proven that it can stop the network degradation, and guarantee the very deep NN works [69].

Then the loss function for DNN is defined

$$\text{Loss}(x, \theta) = \frac{1}{M} \sum_{i=1}^M \|y_i(x, \theta) - \hat{y}_i(x, \theta)\|^2 \quad (35)$$

where the x and θ are the inputs and parameters, respectively and y_i is the sampled output and $\hat{y}_i$ is the output of NN. Then an optimizer can optimize the parameters in NN, to make the loss function as small as possible.

Tripathy & Bilionis [65] use the DNN method as the surrogate to compute the uncertainty propagation. The most concerning question is how many of samples should be used in constructing the NN. From their report, 500 samples can catch enough accuracy of an elliptic PDE on the 2-d unit square domain. And the final UQ result collapse well on the result of MC method.

Actually, the number of 500 is also too large for the scramjet simulation. For decreasing the samples by a better structure, PiNN and TgNN, which are similar, are proposed by introducing the information of differential equations into the NN. As a result, the needed number of samples for constructing such NNs significantly decreases.

Wang et al. [67] reported the usage of TgNN as surrogate for UQ. The core idea of PINN and TgNN is to introduce physical constraints as prior knowledge into the NN training. Considering the generalized differential function

$$u_t + \mathcal{N}(u) = 0, \quad (36)$$

let output $\hat{u}_i$ of NN be the surrogate of field variable $u(x, t)$. Different from loss function (35) in DNN, the loss function in PINN and TgNN read

$$\text{Loss}_u(x, \theta) = \frac{1}{M} \sum_{i=1}^M \hat{u}_i + \mathcal{N}(\hat{u}_i) \quad (37)$$

Similar to the field variable loss function, the boundary and initial condition can be added to generate Loss_b and Loss_i , and the final loss function can be constructed with a convex combination

$$\text{Loss}_{final} = \lambda_u \text{Loss}_u + \lambda_b \text{Loss}_b + \lambda_i \text{Loss}_i \quad (38)$$

The work of Wang et al. proved 30 samples of a 2D transient saturated flow can achieve a satisfactory accuracy. Which is a significant improvement compared with DNN method.

DeepONets is similar to PINN and TgNN except constructing another trunk net. Lu et al. [68] tested its accuracy for a nonlinear diffusion-reaction PDE (see problem 4), and get an accurate surrogate model with 100 samples. Based on this successful surrogate, Psaros et al. [70] made a comprehensive investigation of UQ in PINN and DeepONets.

2.3. Inverse problem

The method described above is the uncertainty propagation method, which means calculating the uncertainty shape of y of mapping (1) when knowing the uncertainty source and its shape. It is an essential part of UQ. Moreover, there is another part of UQ. That is how to confirm the uncertainty of the model parameters through the observed value obtained from the limited working condition experiments to be

applied to the computation of working conditions of a broader range. This is also known as the *Inverse Problem*:

$$\theta = f_{\theta}^{-1}(y^{obs}; x) \quad (39)$$

Usually, the parametric inverse function of mapping f is nonexistent, abnormal or not computable. So solving Eq. (39) directly is impractical. A common alternative is to solve equation [27,71–73] based on Bayesian inferences.

For Bayesian inferences, we need a priori pdf $p(\theta)$ to predict target θ , and likelihood function $p(y_{obs}|\theta)$. The likelihood function is usually written as $L(\theta|y_{obs})$. Thus the posteriori estimate $p^*(\theta|y_{obs})$ can be obtained by utilizing Bayesian formula:

$$p^*(\theta|y^{obs}) = \frac{p(\theta)p(y^{obs}|\theta)}{\int p(\theta)p(y^{obs}|\theta)d\theta} \quad (40)$$

In Section 2.1.1, we introduce the idea of MCMC sampling method. Using the MH algorithm, we regard the posteriori distribution $p^*(\theta|y^{obs}) \propto p(\theta)p(y^{obs}|\theta)$ as the target equilibrium distribution to be solved, and choose the proper transfer probability function $q(\theta^*|\theta)$, such as random walking or independent sampling. Then we can get sample points of the posteriori distribution $p^*(\theta|y^{obs})$ based on Markov chain transfer rule. Since the sampling process gradually approaches the target distribution, we often abandon the initial part of the sample points and retain the sample from the latter part to represent the uncertainty distribution of the target. If there are enough samples, we can restore posterior probability density distribution function using regression. More Bayesian inference methods based on MCMC can be found in Ref. [74].

It can be known from the above analysis that a good choice of priori estimate and likelihood function can lead to a faster approach to target posterior distribution and help reduce sampling calculation. Usually priori estimate $p(\theta)$ can be based on expert opinion or normal distribution using coupling.

$$p(\theta) = ((2\pi)^n \|\Sigma_{\theta}\|)^{1/2} \exp\left(-\frac{\theta^T \Sigma_{\theta}^{-1} \theta}{2}\right) \quad (41)$$

Σ_{θ} is the incidence matrix of component θ of the coefficient vector. For likelihood function, Sargsyan et al. [73] summarize three construction methods of likelihood function, and two method to approximate likelihood function in approximate Bayesian form:

- Complete likelihood function

$$p(y^{obs}|\theta) = \pi_f(y^{obs}; \theta) \quad (42)$$

- Marginal likelihood function

$$p(y^{obs}|\theta) = \prod_{i=1}^N \pi_{f_i}(y_i^{obs}; \theta) \quad (43)$$

- Marginal likelihood function with gaussian approximation

$$p(y^{obs}|\theta) = \frac{1}{(2\pi)^{N/2}} \prod_{i=1}^N \frac{1}{\sigma_i(\theta)} \exp\left(-\frac{(\mu_i(\theta) - y_i^{obs})^2}{2\sigma_i^2(\theta)}\right) \quad (44)$$

N represents the number of measurement of (y_i^{obs}, x_i) means the exact pdf; Mean value $\mu_i(\theta) = \mu(f(x_i; \theta))$ and variance $\sigma_i(\theta) = \sigma(f(x_i; \theta))$ represents statistics predicted under fixed input condition x_i . It is noteworthy that the former two methods need a full calculation of the distribution pattern of likelihood function, which is often difficult to achieve; the latter is more simple and practical for it only needs two type of matrix information. However, for Marginal likelihood function $\sigma_i(\theta)$ with gaussian approximation, sometimes approaches zero gradually as the nature of the data changes, which causes the singularity to appear. So we introduce the approximate Bayesian computing method.

- Approximate Bayesian computation (matching the average value)

$$p(y^{obs}|\theta) = \frac{1}{\epsilon \sqrt{2\pi}} \prod_{i=1}^N \exp\left(-\frac{(\mu_i(\theta) - y_i^{obs})^2}{2\epsilon^2}\right) \quad (45)$$

- Approximate Bayesian computation (matching the average value and variance)

$$p(y^{obs}|\theta) = \frac{1}{\epsilon \sqrt{2\pi}} \prod_{i=1}^N \exp\left(-\frac{(\mu_i(\theta) - y_i^{obs})^2 + (\sigma_i(\theta) - \gamma|\mu_i(\theta) - y_i^{obs}|)^2}{2\epsilon^2}\right) \quad (46)$$

ϵ represents the admissible parameter of approximate Bayesian computation; λ represents the weighting parameter of matching the average value and the variance. Approximate Bayesian computation can be viewed as an approximate to likelihood function and be applied in Bayesian inference process. Its remarkable characteristic is that there is no singularity, which guarantee the robustness of the calculation.

Besides solving inverse problem (40) with the MCMC method, another noteworthy approach is Method of Uncertainty Minimization using PCE (MUM-PCE), which is put forward by Sheen et al. [75] In this method, the pdf of model parameters is expanded using polynomial chaos. As the polynomial coefficient can be viewed as the hyper-parameter of the system, we can optimize the coefficient by employing the Solution Mapping idea of deterministic method to achieve optimal matching of observed results and predicted results. Thus the optimized hyper-parameter determine the uncertainty measure of model parameters. Its detail application cases can be seen in Ref. [76,77].

The methods mentioned above are just standard methods used in engineering. However, solving the inverse problem is another big picture, especially in deep learning. For a more comprehensive insight into this topic, we recommend readers refer to [16].

2.4. Summary of UQ methods

In order to address the complex needs of engineering in the highly intricate and irregular system of scramjet combustion, this chapter presents a wide range of UQ (UQ) methods.

When dealing with less information on uncertainties, the sample method is often the only viable option for detecting uncertainty, followed by the PCE method, which is optimal once the measure of uncertainty has been obtained. However, in many real-life situations, accurate uncertainty measures are not available, and probability intervals can be used as a compromise.

For grid-based models, the MLMC method can significantly accelerate the UQ computations, while PCE-based and collocation methods are suitable for regular mapping systems such as chemical kinetics. In the worst-case scenario, the NN surrogate can be used as the final solution.

Inverse problems rely on uncertainty propagation, and Bayesian inference plays a major role in UQ inverse problems. The choice of likelihood function is a critical task in Bayesian inference.

3. From error analysis and UQ

Error analysis is a traditional topic when dealing with data in engineering, which computing the error propagation through system. Similarly, UQ are concerning the uncertainty propagation through system. So they are sometimes mixed up due to their similar objectives and data approach. Actually, a brief review of error analysis can help engineers understand the UQ tool quickly, and based on this point, a comparison of these two tools is made first.

Moffat [79] gave a brief definition of uncertainty based on error: the uncertainty attributed to a measurement is an estimate of the possible residual error in that measurement after all proposed corrections have been made. In other words, uncertainty is a special error that people cannot reduce anymore. This definition is confusing because it is hard to judge whether the error can be reduced, especially with the developing measurement tools. Taylor [80] stated them more explicitly by a mathematical expression. Generally, the data obtained by experiment measurement is expressed as:

$$x = x_{\text{best}} \pm \delta x, \quad (47)$$

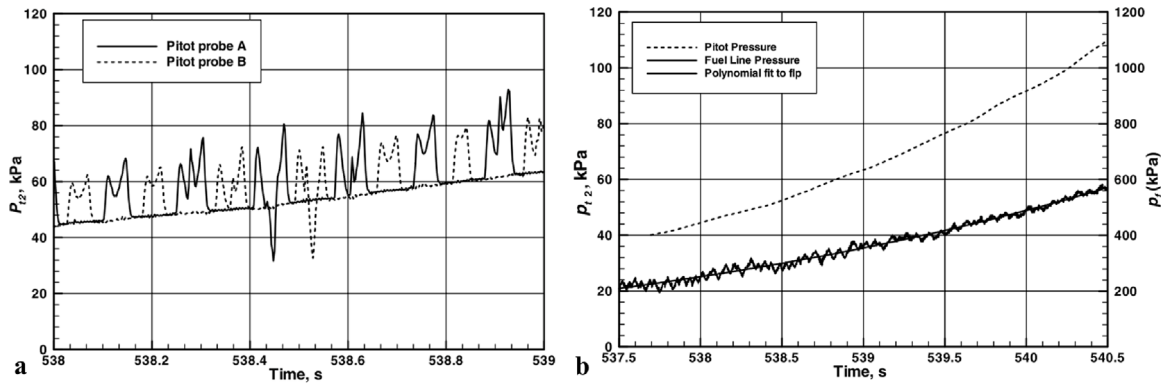


Fig. 4. Pitot probe history; b. Pressure used for analysis [78].

where x_{best} is the best estimation and δx is the uncertainty in error analysis. Then the real quantity is believed to lie within the range $x_{\text{best}} \pm \delta x$ with a certain percent confidence, 95% for example. This statement does not distinguish uncertainty and error and expresses them by a scalar. As a result, a typical uncertainty propagation of mapping $y_{\text{best}} \pm \delta y = f(x_{\text{best}} \pm \delta x)$ in error analysis read

$$\delta y = \left| \frac{df}{dx} \right| \delta x. \quad (48)$$

This definition is also utilized by Coleman&Steele [81], and they called this process the uncertainty analysis based on *Guide to the Expression of Uncertainty in Measurement*. Eq. (48) is simple but should be under the condition that δx is small enough. If not, it would miscalculate the confidence of quantity y within $y_{\text{best}} \pm \delta y$.

For now, people can describe x in more detail in a statistical manner, which means it is not just about the range, but also the measure of the function $x(\xi)$. So in the UQ process, people are more concerning the complete mapping process of statistic variables, including the range and measure. As a comparison, error analysis solves scalar δy , and UQ solves the statistical variable of $y(\xi)$ for mapping $y(\xi) = f(x(\xi))$. The critical difference between UQ from error analysis is the representation of $x(\xi)$, which the former mainly relies on statistical characteristics. The cost of mapping $f(\cdot)$ is another concerning problem for error analysis and UQ. Error analysis needs few samples, and UQ relies on “surrogate” for decreasing numerical computing time, where the “surrogate” means certain patterns in the mapping m in Eq. (1). Even though, for highly complicated problems, the simple boundness δy is also useful [82].

Based on Moffat’s and Taylor’s work, we know UQ is an extension of error analysis. Limited by the mathematical tool and computing power in the past, researchers only dealt with some simplified uncertainty and operators, which is called error analysis. Then with the development of mathematics and cluster, the statistical variables $x(\xi)$ can be represented and computed more efficiently based on the advanced sampling method and generalized PCE. It opens the door to a closer approach to the real, uncertain world and thus offers more detailed information for engineering design. Some recent applications, like chemical kinetic model reduction, and turbulence calibration, demonstrate the benefits of considering UQ. Furthermore, people turn to the scramjet combustor, a more complicated problem with a high dimension of uncertain source.

4. Applications of UQ in scramjet

UQ method is still in the early stage of development. For the high-dimensional, less regular problems in a real environment, though the existing sparse construction has been employed, the considerable computing load still restricts its direct application in the numerical simulation of the complete engine. In most cases, UQ is applied in some subsystems of the engine combustion system, including the UQ of the chemical kinetic and turbulence models. However, some researchers

still attempt to apply UQ to the simulation of the complete engine [83–85]. Although these efforts are at very first stage, they still have made some exciting achievements. In this chapter, we will introduce the related work.

4.1. Chemical kinetic models

The most important model of scramjet simulation is the chemical kinetics model. For a complete scramjet simulation, the time cost in solving chemical kinetics may occupy above 50% of total time cost. In most situations, the single-step mechanism is the only choice [86,87]. So a detailed review of UQ application in chemical kinetic models is made first.

4.1.1. Uncertainties in models

Considering the production rate in chemical kinetics

$$\dot{\omega}_i = W_i \sum_{j=1}^{nr} [(v''_{i,j} - v'_{i,j})(k_{fj} \prod_l^{ns} (\frac{\rho z_l}{W_l})^{v'_{l,j}} - k_{bj} \prod_l^{ns} (\frac{\rho z_l}{W_l})^{v''_{l,j}})], \quad i = 1, \dots, ns \quad (49)$$

where W_i is molecular mass, $v'_{i,j}$ and $v''_{i,j}$ are the forward reaction chemical equivalent coefficients and the backward reaction chemical equivalent coefficients, respectively. k_f and k_b are, respectively, the forward reaction rate constant and the backward reaction rate constant, where k_b can be considered as a function of k_f , $k_b = k_b(k_f)$, so it is excluded from consideration for clarity. By default, k_f of each elementary reaction remains unchanged in any reaction, which is the rate criterion. Therefore, determining the reaction rate constant of an elementary reaction is the core problem of the combustion dynamics model. Through further modeling of k_f , the Arrhenius equation can be obtained:

$$k_f = AT^n \exp(E_a/RT) \quad (50)$$

where T is the temperature. The parameters (A, n, E_a) are to be determined by experiment data k_f , and a regression process is included. Since there is always a scattering phenomenon in the experimental data of k_f , even under the assumption of the high-fidelity of the model, the regressed parameters (A, n, E_a) are only an approximation and unavoidably contain uncertainty.

Take the primary elementary reaction in the hydrogen–oxygen mechanism model as an example: $\text{H} + \text{O}_2 \leftrightarrow \text{O} + \text{OH}$ [88]. After experimental measurement, the fitted formula parameters are shown in Table 1. From this table, we can find the best result of pre-factor A remains around 3% uncertainty without any knowledge of the uncertainty measure. Besides the uncertainty introduced by the scattering experiment data, the uncertainty introduced by modeling is also critical. The theoretical Arrhenius equation is a two-parameter equation, $k_1 = A \exp(E_a/RT)$. Then for collapsing to experiment data, it is

Table 1

Low temperature reaction rate coefficient k of elementary reaction $H + O_2 \leftrightarrow O + OH$. The parameters are uncertain [88].

A ($\text{cm}^3 \text{ mol}^{-1} \text{ s}^{-1} \text{ K}^{-n}$)	n	E_a/R (K)	T range (K)	Reference
$(1.68 \pm 0.19) \times 10^{14}$	0	8119 ± 139	960–1710	[90]
$(6.93 \pm 0.96) \times 10^{13}$	0	6917 ± 193	1103–2055	[91]
1.59×10^{17}	−0.927	8493	1050–2700	[92]
$(7.13 \pm 0.31) \times 10^{13}$	0	7065 ± 140	1050–2500	[93]
6.73×10^{15}	−0.50	8390	950–3100	[94]
$(1.12 \pm 0.08) \times 10^{14}$	0	7805 ± 90	1100–1527	[88]
$(1.04 \pm 0.03) \times 10^{14}$	0	7705 ± 40	1100–3370	[88]+[95]

modified by adding the term T^n , which becomes the commonly used three-parameter equation. The modified Arrhenius equation cannot accurately simulate the ignition delay time at low temperatures [89]. The above facts show that the Arrhenius equation cannot accurately reflect the complex physical process in chemical reactions, and using a three-parameter equation for fitting will inevitably bring about epistemic uncertainty.

4.1.2. Uncertainty calibration of model parameters

The chemical reaction model is built based on the production rate similarity, which means elementary reaction has similar production rate in different chemical reaction processes. So completely unknown chemical reaction model can be built upon known elementary reactions. This indicates that how to model the chemical reaction uncertainty plays a decisive role in simulating the real physical process. Since the production rate is obtained by fitting the Arrhenius formula, it can be further concluded that the uncertainty of the production rate is decided by the coefficients of the Arrhenius formula [96].

Zsély et al. [97] conducted uncertainty analysis on kinetic parameters of a combustion system of hydrogen/air and carbon monoxide/air using local and global UQ. The MC method with acceleration by Latin Hypercube Sampling is used. The author employed local UQ and experimental data to obtain mean square error σ of uncertainty parameters, and assumed that the uncertainty shape of the parameters is pseudo-normal distribution after 3σ truncation. The study ignored the covariant uncertainty between the parameters. The computational result indicates that the uncertainty of ignition delay time is small for both kinds of fuel, which basically matches scatter of experimental data.

Sheen & Wang [98] conducted research on the experimental data [99] of the N-heptane oxidation process using MUM-PCE. The authors assume that the covariance exists between parameters of one elementary reaction but do not exist between different elementary reactions. They prove that high precision data for components measurement is a strong restriction for UQ, which provides a high-fidelity computational model with calibrated model parameters. So the authors proposed that the global flame property measurement should keep up with the accuracy of model parameters to ensure further improvement in model accuracy. More studies [100–102] found that the model's predictive ability can be further enhanced by considering the existence of covariant relation between elementary reactions in different chemical reactions. This phenomenon may go against the independence of elementary reactions, which is the base of construction of detailed mechanism. Therefore, for the covariant relationship between different elementary reactions, it deserves further study for its physical meaning.

4.1.3. Influence of model parameters uncertainty

Based on uncertainty information, we can define the sensitivity of different elementary reactions on the value we concern with, and thereby we can rank the importance of the reactions. Valkó [103] defines the global sensitivity measurement using HDMR decomposition with Rosenblatt transformation. The definition is similar to Sobol decomposition [104]. The author studied the wall-loss reactions at the explosion limit of hydrogen/oxygen reaction. They found that the

correlation sensitivity dominated the variance of all output results except the wall-loss rates. However, it should be noted that correlation sensitivity does not represent the importance of model parameters and independent correlation sensitivity play a more important role in affecting the analysis. Valkó [105] conducted further research on the combustion process of synthesis gas/air and expressed the same opinions as paper [103]. However, after comparing the global sensitivity analysis with the local sensitivity analysis, the author believes that the local sensitivity analysis is an accurate enough approximation method for this kind of reaction and is more efficient. Tao et al. [106] defined the influence factor with the local sensitivity and studied the importance of elementary reactions in hydrogen, hydrogen/monoxide, methane, methanol, formaldehyde and ethylene. The authors concluded that UQ optimization can dramatically reduce the uncertainty in the prediction of ignition delay. They also indicate that the uncertainty information of critical elementary reactions would play a vital role in the prediction results if controlled within 20%.

4.1.4. Model simplification

The complete chemical kinetics model includes enormous reaction components and elementary reactions. Using the detailed kinetics model in scramjet simulation is impossible. One problem is that the computational load is unaffordable. Another one is that the time scale of different reaction models has great differences, which causes strong stiffness of the system and easy computational divergence. So it is an important issue for the chemical reaction kinetics model to develop a simplified model, with less stiffness and high fidelity [107]. The model simplification method based on UQ benefits the fidelity assessment of UQ methods. The larger uncertainty exist in parameters, the lower-fidelity the model is. So it is easy to find a high-fidelity model in simplified models. Xin et al. [108] simplified the model of normal butane and iso-butane mechanism using directed relation graph (DRG) and the matched sensitivity analysis. MUM-PCE was applied in the parameters reduction and restriction of the simplified model. The authors find that the dependence of model uncertainty is subject to the completeness of the model [108]. On the other words, once the successive simplified models contain the most important reactions, the uncertainty should be decreased respect to increasing the model size. For the incomplete models, the behaviors of uncertainty are unpredictable. Generally speaking, if the removed reactions and the remaining reactions have the property of second-order coupling, the simplified reactions can be regarded as incomplete. Recently, Malpica Gallassi et al. [109] developed a solution of model simplification based on UQ. This solution employs the analytical method of computing singular perturbation as a certainty tool to obtain marginal probabilities of elementary reactions; when the marginal probability of one elementary reaction is higher than the given threshold value, this elementary reaction will be put into the simplified model. By simplifying the n-butane model, the authors give out the asymptotic behavior of accuracy to the threshold value, which proves the method's reliability.

4.2. Scramjet

4.2.1. Uncertainty in the inflow of combustor

In the engine experiments, uncertainty in physical parameters is expected due to the limited resolution of measurement devices and the modeling of imperfect data. To be closer to this uncertainty, a brief review of Hyshot-II experiments is present and highlights the uncertainty source of the inflow parameters, which the numerical simulation would use.

Hyshot-II is a scramjet prototype successfully tested by ground experiments and flight. Smart et al. [78] reported the comparison of the flight data and the ground data of the HyShot-II, where the flight took place at Woomera Prohibited Area Test Range in central Australia, with an unguided sounding rocket. The ground experiments were conducted by the Center for Hypersonics at the University of Queensland with

Table 2

Part of experimental conditions from HyShot II ground tests performed in the HEG [110].

	Run	P_0 [bar]	T_0 [K]	H_0 [MJ/kg]	P_{H_2} [bar]	ϕ
Fuel-off	805	178.05	2742	3.25		
	807	181.19	2777	3.30		
	808	175.27	2716	3.21		
Fuel-on	804	172.96	2652	3.22	5.41	0.341
	810	176.66	2705	3.20	5.73	0.351
	812	179.84	2729	3.23	5.32	0.325
	817	176.99	2796	3.13	5.28	0.315
	828	187.43	2796	3.28	4.84	0.288

shock tunnels. There is a 3 s flight test window between 537.5 ~ 540.5 s, where the height is around 29 ~ 35 km. The Mach number of the flight is around 7 ~ 8. Due to the failure of the radar's tracking, the flight trajectory is reconstructed by the data of the loaded measurement devices.

Fig. 4-a presents the pitot pressure for the inflow, and Fig. 4-b presents the modeling of pitot pressure and the fuel line pressure. The pitot pressure has a systematic uncertainty of ± 0.5 kPa due to the roll and nutation of the payload. Therefore, for available data for comparison and computation, the average of pitot pressure A and B is applied in Fig. 4-b for excluding the excursion. Unfortunately, this postprocessing makes the uncertainty in pitot pressure data more complicated.

On the other hand, the fuel line pressure has an obvious uncertainty of ± 6.9 kPa in Fig. 4-b. However, when the polynomial fitted line is used in a deterministic numerical solver, this important uncertainty is neglected, and it may lead to a less robust design.

Constantine et al. [110] presented more details of the ground test data which is obtained by the High Enthalpy shock tunnel Göttingen(HEG) of the German Aerospace Center(DLR) [111], shown in Table 2. This experiment measured the stagnation pressure P_0 , temperature T_0 , and enthalpy H_0 of the nozzle supply region. In addition, the hydrogen fuel plenum pressure P_{H_2} is also measured for fueled runs. As shown in Table 2, which is a typical experiment record, the obtained data is different from each other runs. This independent statistics is different from the time-correlated statistics in one run, and is high quality but has less samples. To represent this statistics, the inflow parameters should be modeled with uncertainty.

4.2.2. Uncertainties in the geometry of the combustor

Another source of uncertainty in the combustor is the surface roughness due to the extreme working condition. The highest combustion temperature can be around 2000 K, which causes severe thermal load on the surface of the combustor walls. Uhrig&Larrie [112] investigated the possible damage on the composite material C/SiC for scramjet. In the experiment, the composite pilot endured the temperature of 2200 K, which recovered the flight condition $Ma = 7$. After 3 thermal cycles, the pilot looked unscathed, but the thermal insulation was lost, as shown in Fig. 5. The experiment validates the durability of composite material for the operating condition of the scramjet, but the lost thermal insulation make a visible rough surface of the pilot, and introduce a geometrical uncertainty.

So far, little literature reports this uncertain roughness's influence on combustion. However, the flow variation induced by regular roughness has been observed for a long time. Ito et al. [113] and Zhou et al. [114] show the transition mechanism by different roughness and validate the enhancement of the transition process by the roughness. Ekoto et al. [115] got closer to the details of stress profiles of supersonic boundary layers with square and diamond topology. Compared with a smooth surface, the turbulence induced by diamond roughness deviates is much more complicated than that induced by square roughness because the diamond roughness produces significant localized distortions. The above work illustrates the importance of research on the uncertain roughness, which the experiments can observe.

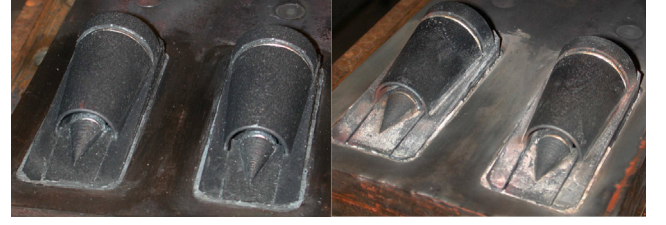


Fig. 5. Left: pilot before test; right: pilot after 3 cycles [112].

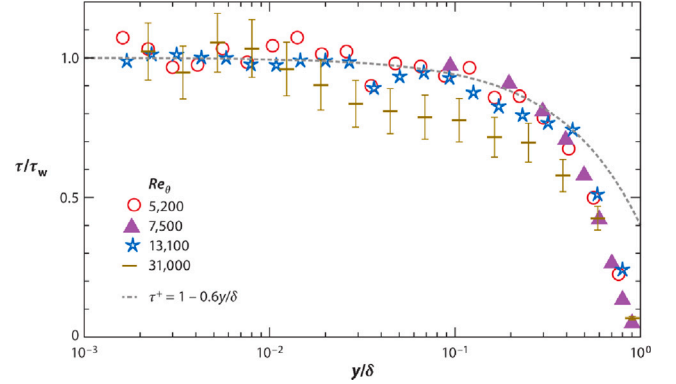


Fig. 6. Total stress(sum of viscous and Reynolds stress) versus wall-normal distance [117]: Red circles: $Re_\theta = 5200$ [116]; Purple triangles: $Re_\theta = 7500$ [118]; Blue stars: $Re_\theta = 13100$ [116]; Dark yellow lines: $Re_\theta = 31000$ [116]; Dashed gray line: $\tau^+ = 1 - 0.6y/\delta$ [119]. An estimation of the uncertainty is shown for $Re_\theta = 31000$.

4.2.3. Turbulence model

The uncertainty in the turbulence model has attracted much attention in recent years. The necessity of introducing uncertainty comes from two facts: one is the uncertainty of experiment data, also called aleatory uncertainty, and another is the models with various fidelity, which is named limited epistemic uncertainty.

De Graaff&Eaton [116] examined the Reynolds stress for a flat-plate turbulent boundary layer from $Re_\theta = 1430$ to 31000 with a laser-Doppler anemometer. 1.5% and 10% of uncertainty are reported in the center of the profile for mean velocity U and Reynolds stress $u'v'$, respectively, and they decrease to zero at the wall. However, the accuracy in predicting the stress along wall-normal distance is critical to the success of the turbulence models. Once the uncertain experiment data is used to validate the models, the need for UQ is unavoidable. Specifically, Bose&Park [117] reviewed the turbulence models for near-wall eddies in high-Reynolds number boundary layers based on large-eddy simulation and summarized the data from DeGraaff&Eaton, which is present in Fig. 6. The figure reports the uncertainty with a dark yellow vertical bar, which represent the confidence interval, for the case of $Re_\theta = 31000$. The theoretical line of dashed gray does not fall in the range of confidence interval, which implies the model fails at this Re number. On the other hand, the uncertainty in the experiments data challenge the modeling in a deterministic way, if there is no concrete theoretical support.

Secondly, the inaccuracy of low-fidelity models is generally considered another uncertainty source. For example, many Reynolds Averaged Navier Stokes(RANS) models rely on the Boussinesq hypothesis

$$\overline{u'_i u'_j} = \frac{2}{3} k \delta_{ij} - 2\nu_t S_{ij} \quad (51)$$

where u'_i is the i th component of velocity fluctuation, k is the turbulent kinetic energy, ν_t is the eddy viscosity, and $S_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})$ is the mean strain rate tensor. However, this hypothesis is only strictly correct for inflows in which the turbulence time scale is much smaller than the

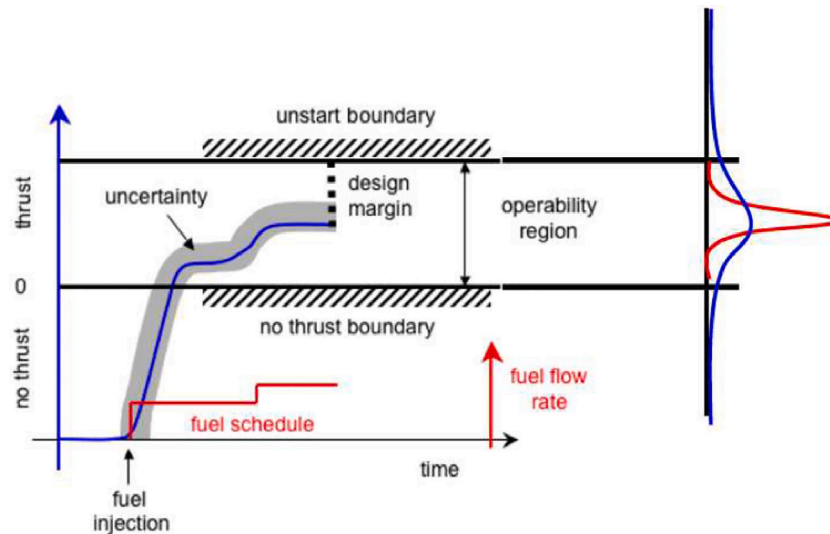


Fig. 7. Sketch of the performance envelope and uncertainty of scramjet Hyshot II [83].

mean shear time scale. Otherwise, the Reynolds stress will be subject to non-local effects [120]. This low-fidelity model, compared with high-fidelity model like Reynolds Stress Transport Models(RSTM), introduce the epistemic uncertainty.

4.2.4. Application in hyshot and hifire program

In recent years, the scramjet above Mach 5 has drawn wide attention in UQ [121]. Unlike the ramjet, its work environment of the combustor is supersonic flow. Though the geometric environment is rather simple, the large Reynolds number, complicated flow structure and huge coupled stiff chemical reactions pose a great challenge to the knowledge of the physical process and numerical computation. For public area, Hyshot program and HIFiRE program are the most famous scramjet projects. So most public researches of scramjet are based on the configurations from these two programs.

Witteveen et al. [83] introduce the influence process of the uncertainty of the fuel injection process on engine performance envelope(see Fig. 7). Combined with Simplex Stochastic Collocation(SSC), an UQ solution was developed for the quantitative analysis of the random uncertainty, subjective uncertainty and numerical errors. They studied the experimental data and uncertainty spreading process of scramjet HyShot-II [2,78] and concluded: the 99% confidence interval of the wall pressure result has an uncertainty of 8.026%, in which the subjective uncertainty contributes most to the uncertainty of the final result. To balance the contribution of different uncertainty resources, the authors proposed to change the fidelity of the computational model to reduce the contribution of the subjective uncertainty, for example, replacing the RANS solver with the LES solver. Crow [122] researched the uncertainty problem of thermal radiation calculation of scramjet Hyshot-II and HyFire-2 [123,124]. In the computation, the authors used the Discrete Ordinates Method (DOM), which involves molecular spectral model and grid refinement. So they mainly studied the subjective uncertainty of parameters in the two numerical processes. By adding the uncertainty into the computation as the linear perturbation, the authors found that the thermal radiation calculated by DOM has 5% uncertainty.

For HyShot-II, Constantine et al. [110] introduce activated subspace method in UQ. The authors indicate that a complete numerical simulation of the model needs 9500 CPU hours, so the traditional response surface method is difficult to realize for the uncertain source problem of seven-dimension considered in this paper. The basic idea of the activated subspace method is to give the most important direction of uncertain source space using linear fitting and calculate the uncertainty in the activated subspace to reduce the computational effort. It can be

seen from the results that the important influence factors of outlet pressure are attack angle, total temperature, total pressure and turbulence intensity. The lesser important factors include the turbulence scale, the turning point of the fairing and the wedge. A more important finding is that the changing of outlet pressure in the obtained important direction exists a good linear correlation (see the Fig. 8). In this way, a seven-dimensional problem can be degraded to a one-dimensional uncertainty problem.

Geraci et al. [125] introduce multifidelity multilevel Monte Carlo method(MFMLMC) into the uncertainty analysis of HIFiRE scramjet, which is the generalized version of MLMC. Huan et al. [6,84] follow Geraci's work and try to find the sparse structure of multidimensional uncertain space by combining the MFMLMC method, polynomial chaos expansion, and compressed sensing method for the purpose of obtaining the most economical UQ method. With the integrated UQ scheme, the authors studied the possibility of covering the prediction results of the high-fidelity model by calibrating the parameter uncertainty of the low-fidelity model, as shown in Fig. 9. The results indicate that by retaining the uncertainty of constant C_R in the static Smagorinsky model, the predicting results of the dynamic Smagorinsky model can be covered well; however, by retaining the uncertainty of some parameters in two-dimensional LES cases, the computed results of three-dimensional LES cases can hardly be covered. The authors think that the former different fidelity models does not cause a qualitative change in the flow field, so it can be successful. While the latter pair of models lead to different physical processes. For example, the bow shock of a 3-D LES case cannot exist in a 2-D LES case. Thus the prediction fails.

Geraci et al. [85] tried the scramjet design optimization under uncertainty. They use the MLMC method as the kernel of optimization process. In the test model, they achieved an overwhelming advantage over the MC method and proved the distinguishing difference between optimizations with and without uncertainty. However, when extending the framework to the near-real scramjet, the incredible number of LES realizations is still a big problem for UQ. They fails having an explicit conclusion about the optimization with uncertainty.

5. Conclusion and future study

This review summarizes the available UQ methods for the scramjet simulation, which is a high-dimensional, non-linear, less regular and computationally intensive system. All of the properties challenge the limits of various UQ methods. Then the uncertainty sources in scramjet are explored, and the attempts to deal with these uncertainty are present. The summary of the work of UQ in scramjet is as following:

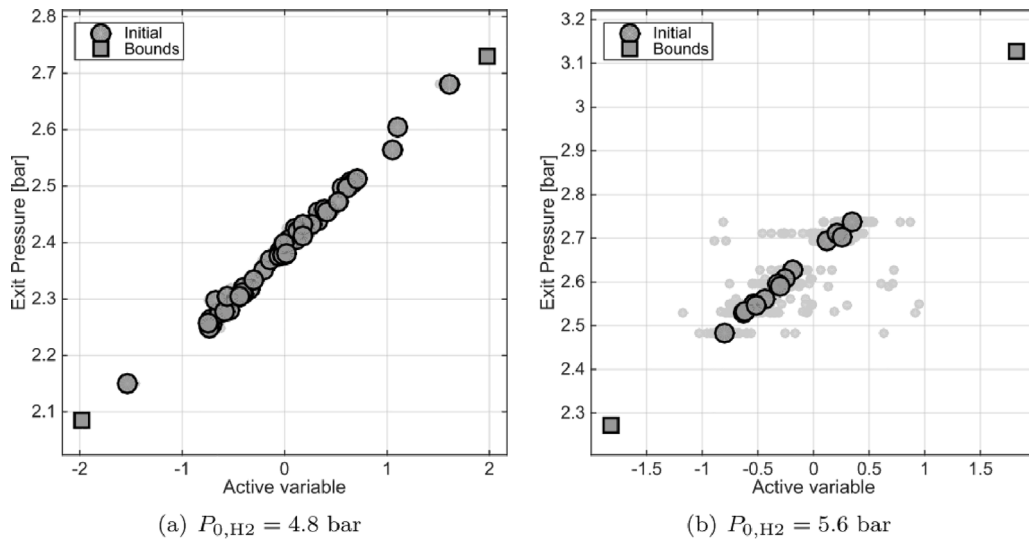


Fig. 8. In the uncertainty source space of 7 dimension, the computational results of Constantine et al. show that the uncertainty of outlet pressure is monotonic in one direction, under conditions: $P_{0,H_2} = 4.8$ bar (left) $P_{0,H_2} = 5.6$ bar (right) [110].

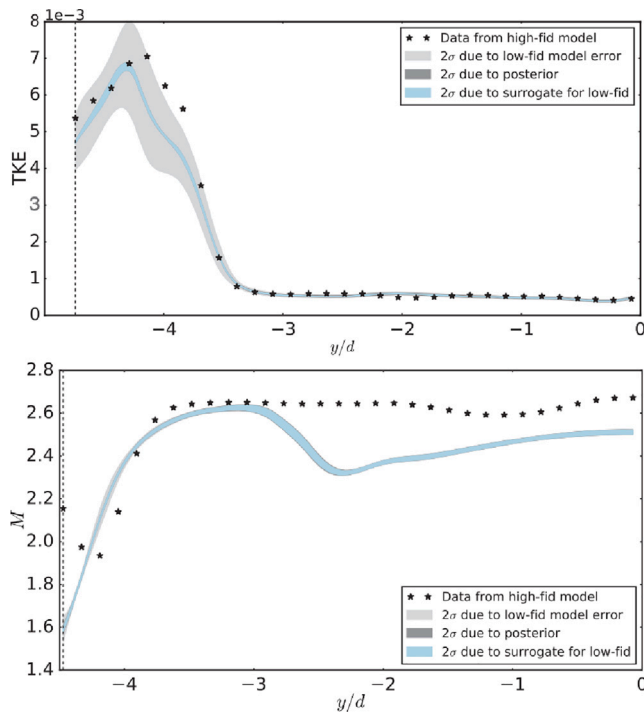


Fig. 9. The prediction of low-fidelity model with uncertainty covers the prediction of high-fidelity model. left: results of TKE based on static Smagorinsky model(low-fidelity) and dynamic Smagorinsky model(high-fidelity); right: results of Ma based on 2-D configuration(low-fidelity) and 3-D configuration(high-fidelity).

- The probability theory and corresponding numerical tools are prospering under the demand of dealing with uncertainty. Since 2000, many applied mathematicians and engineers have put a lot of effort into developing the basic theory and numerical tools of UQ methods. These methods includes various Monte Carlo methods, variants of the PCE methods, collocation methods, and neural network surrogate models. On the other hand, The Bayesian approach dominates inverse problem. However for now, there is no a mature solution for the high-dimensional, nonlinear and less regular system of the combustor. Considering the unavoidable uncertainties in scramjet, there is a long-term demand for state-of-art UQ methods.

- The working environment of engine combustion chamber is complex and the uncertainty in the simulation is unavoidable. The material of combustor suffers extreme conditions and accumulates the damages. The inflow from the compressor, the geometrical configuration and the manufacturing errors of the combustor are hard to control, and finally become the sources of uncertainties. In the process of numerical simulation, facing flow structures and chemical reaction processes that span six or seven orders of magnitude, finite computational resources inevitably encounter under-resolved scales. These under-resolved scales are finally reflected in uncertainties of various mathematical model parameters. Therefore, for the accurate and robust evaluation of the real working condition of the engine, the numerical tools focusing on the engine combustion computation must be incorporated in the UQ.
- There always are some researchers insist incorporating UQ in the scramjet simulation. Many researchers have already tried various UQ tools into the combustion and engine combustor simulation, which proves that the UQ already has certain possibility under the current computing resource. The researchers have primarily investigated the uncertainty source and the propagation for certain parameters, such as the engine's wall pressure, outlet pressure and thermal radiation [122]. These are important data for future engine design.

In the face of these realities, we make the following suggestions for the future work.

- ★ Intrusive Galerkin method plays an important role in the mathematical analysis of elliptic equations and Boltzmann equations, and it is challenging to implement this method in hyperbolic equations [48–50,127–129]. They also have not shown obvious advantages in complex engineering problems. Especially for a scramjet engine, which is composed of many modules, the complexity of each module and their interactions make it difficult to apply the reformulation approach. A possible solution is to use programming techniques such as C++ operator overloading and template deduction to implement the entire work, similar to the automatic differentiation programming.
- ★ The core of non-intrusive UQ methods is the sample. UQ methods usually focus on reducing the number of samples while ensuring accuracy, but reducing the time for each sample should also be considered. Methods for reducing the sampling time include using

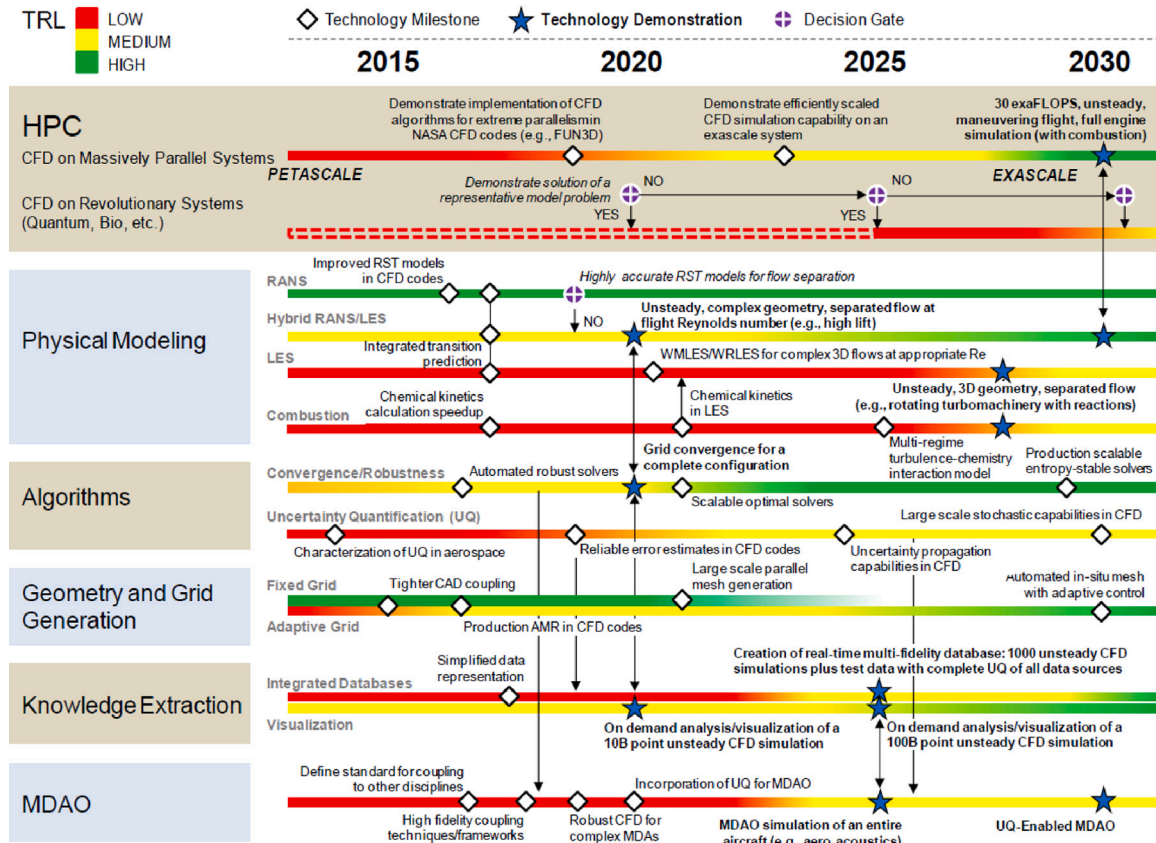


Fig. 10. Development path of CFD [126].

NNs to replace the differential equation models in CFD, such as chemical reaction and turbulence models, and adapting various CFD algorithms to evolving acceleration cards and heterogeneous HPC architectures. Note that CFD is a memory-intensive algorithm, and data locality is the key to adaptation. Although the above two are not UQ methods, they play a crucial role in the development of UQ.

- ★ Back to the UQ methods, the MLMC method is one of the successful methods applied to simplified scramjet calculations and is an important part of UQ. It is a special case of the multi-fidelity methods, but with the aid of grid coarsening, its theoretical performance is optimal. In the actual implementation process, the multi-fidelity methods can avoid the restriction of the grid and has greater flexibility, making it more worthy of attention in the engineering field. The multi-fidelity method is an acceleration technique for UQ, and a suitable UQ algorithm must still be selected as the kernel to complete the entire UQ calculation.
- ★ The Gaussian process is basically ineffective when dealing with extremely complex problems such as scramjets. When the uncertainty is small and the mapping is regular, HDMR or collocation methods should be considered first. Under the above condition, both of polynomial and collocation methods can accurately capture the mapping pattern, and the theoretical minimum number of samples can be obtained. For complex problems such as scramjet engines, they are suitable to be used in local small models, such as the elementary reactions with reaction rate uncertainty factor less than 1.5 in the kinetical model. For less regularity problems, the finite element PCE method can alleviate the accuracy problem of the global PCE.
- ★ In scramjet simulations, when various sub-modules are combined together and the uncertainty is high and the regularity of the mapping is poor, the fitting ability of polynomial and collocation

methods rapidly decreases. At this time, we need to consider using NNs. Once the NN can successfully fit the mapping, then traditional MC methods can handle most UQ problems. However, all networks have no theoretical guarantee and usually require a large number of samples to ensure their fitting ability. The goal of PINN and DeepONets is to adapt to the differential equations patterns to achieve the purpose of training the network with fewer samples. On the other hand, if there is no pattern information accessible in the differential equations, brute force search can be used to try the optimal structure of the DNN.

- ★ Each UQ method has its own advantages and limitations. In the face of the extremely complex problems of scramjet engines, it is necessary to use an appropriate combination of various UQ methods for each local problem, and form a comprehensive UQ solution for scramjet simulation.

Another noteworthy trend not detailed in this article is the development prospect of future CFD, high performance computing(HPC) and their influence on UQ. According to the report [126], the future of CFD development is closely tied to the development of HPC. Fig. 10 illustrates the projected development of UQ in CFD over the next two decades, indicating that the technology will become an important component and will be fully developed as computational power continues to improve. Combining UQ with multi-criteria analysis, multidisciplinary analysis and optimization (MDAO) will be an important focus going forward. HPC centers are currently being constructed, global computing power is increasing significantly, and CFD platforms are being established around the world, all of which create the necessary demand and conditions for the development and application of UQ engineering. Additionally, UQ data can be used to develop a series of numerical tools, such as multi-objective global optimization and global sensitivity analysis, which will greatly improve the global capability of numerical design, accelerate the design loop and facilitate better engine designs.

CRediT authorship contribution statement

Linying Li: Conceptualization, Methodology, Software. **Zhonghua Zheng:** Data curation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

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